You Can’t Outsource Production Judgement

Why AI Makes Unreliable Narrators of Us All

Joe Woodhouse | DevOps Not Dead | London 2026

What We’re Covering Today

Need 90 mins to do this properly.
Could rush it in 60. I have 25.

I’ll go deep on what I can.
The rest is in the appendices.
You’ll get the complete deck.

Act I

TL;DR

Bottom Line Up Front

LLMs are unreliable narrators

… And they make us one, too.

Individual risks vary with the amount of outsourcing

… Counter-intuitively, mid-level is the riskiest.

Risks cascade: individual to department to company.

Why You Don’t Believe Me

We feel faster with genAI LLM.
We believe you’re still doing the cognitive work.

So did they.

Why You Don’t Believe Me, cont.

METR Study (2025), gold-standard RCT

Experienced devs. Real paid work on real repos.
Subjective self-reporting: 24% faster.
Objective measurement: 19% slower.

They didn’t notice a 43% performance gap. Why?

Outsourcing Hierarchy

Level 1: AI Assist, or “Help me as I do it”

Semantic search, syntax completion

Level 2: AI Augment, or “Do some of it for me”

Prompt-by-prompt code gen, debugging, testing

Level 3: AI Outsource, or “Do it, tell me when ready”

Vibe coding, agentic, autonomous workflows

Outsourcing Hierarchy, cont.

Counter-intuitive finding, independent studies:

Level 2 is the most dangerous to individuals.

- We still engaged
- We believe we’re still doing the work
- AI substitutes for our skill while we watch
- Skill decay occurs outside our awareness

Outsourcing Hierarchy, cont.

Level 3 risks are catastrophic…

but nobody’s fooled about contribution

A Touchstone Test

Can we explain the answer and how it was reached
… without going back to check the LLM chat?

Yes: it’s our answer, we learned, we can recall it.

No: We used AI instead of skill.
… and we may have lost something

Act II

“Extraordinary Claims Require Extraordinary Evidence”

The Mechanisms

M1: Unreliable Narration

M2: Skill Decay

(M3: Security/Quality Debt)

(M4: Cognitive Overload Paradox)

(M5: Autonomous Agent Catastrophe)

Unreliable Narration - BLUF

We can’t trust LLM outputs.

“Hallucinations”; non-determinism; GIGO

We can’t trust our own perceptions of LLM use.

Contaminated planning. Metrics. Risk assessment.

This is most dangerous at Level 2: AI Augment.

We don’t notice what’s happening.

Unreliable Narration - Evidence

METR (2025): The 45% Perception Divergence

- Experienced devs, real OSS repos, RCT
- Claimed: 24% faster | Actual: 19% slower
- Perception diverged 43% from reality
- They still believed they were faster post-study!

Unreliable Narration - Evidence

MIT (2025): Cognitive Debt Study

- 54 students, 4 months, writing SAT essays
- LLM users: 83% couldn't recall own essay
- EEG showed alpha/beta waves halved
- Neural engagement declined over time

Unreliable Narration - Evidence

Frontiers in Psychology (2025): Substitutive Offloading

- Users overestimate knowledge of AI output
- LLMs create “illusion of competence”
- Memory encoding & retrieval decline over time
- … User is unaware

Unreliable Narration - Cascade

Harms to individuals harm the Eng department

  Impact of 43% perception gap on estimation

  Project estimates driven by self-reports

Faros AI study of 10,000 devs

  Everyone claimed LLM-driven speedups

  DORA metrics flat

Unreliable Narration - Cascade

Harms to Eng department harm the whole org

- Shipping cadence
- Code quality, code effort
- AI metrics likely disconnected from P&L

We already don’t usually measure what matters

- Consider AI effects on PRs, lines of code, etc.

Unreliable Narration - Mitigations

1. The Touchstone Test
   Explain the working without checking the chat?
2. Attempt the problem first before using AI
   “Desirable difficulties” create durable learning
3. Anthropic Safety Interaction Patterns
   Conceptional questioning = test score 65-86%
   Full delegation = test scores 24-39%

Unreliable Narration - Mitigations

4. No-AI Day(s)
   Compare AI-on performance to AI-off
   Similar tasks
   Track speed, errors, confidence, actual outcomes
5. Measure what matters
   Bugs in PROD, INCs (especially security), rollbacks
   NOT: commits, PRs, lines of code

Unreliable Narration - Mitigations

6. Explanation-based code reviews
   “Walk me through how this works”
   Not just “Does it past CI/CD tests?”
   Forces the dev to demonstrate understanding
7. AI-on/AI-off A/B testing at team level as well
   Measure: bugs, INCs
   Detect if “AI productivity” is eroding quality

Unreliable Narration - Mitigations

8. AI is change, so use change management
   Not: “here’s the tool, now go faster”
   But: “here’s how cognition shifts & how to detect”
9. Anthropic AI Fluency Index Training
   11 behaviours that predict effective AI use
   e.g. questioning outputs leads to better outputs

Unreliable Narration - Mitigations

10. Train your users
    - ManpowerGroup: 56% received zero AI training
    - Include all previous mitigations
    - Show them this presentation :)

That’s All For Now

We’ve covered the most important mechanism.
There are four more in the appendices.

If You Only Remember One Thing

The Touchstone Test is at the heart of this.

Can you explain how the answer was arrived at,
without going back to check the chat?

Yes: You learned, AI scaffolded your learning. Safe.

No: AI substituted for your learning. Not safe.

AI Requires Safety-Critical Thinking

Aviation doesn't ban autopilot.

It mandates minimum unaided practice.

Medicine doesn't ban diagnostic AI.

It mandates clinician benchmarking without AI.

The question isn't "Should we use AI?"

The question is "At which level, for which task, with which safeguards?"

Thank You, and Questions

Questions?

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Living document available at